

NORTH-WEST UNIVERSITY (MAFIKENG CAMPUS)

CMPG 325 – COMPUTER NETWORKS

INDIVIDUAL ASSIGNMENT

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PROJECT ID: CMPG325-2026-064

CLIENT ID: CLI – 064

**Organisation: Motswaledi Civil And Structural
Engineers (Mafikeng)**

Industry: Engineering

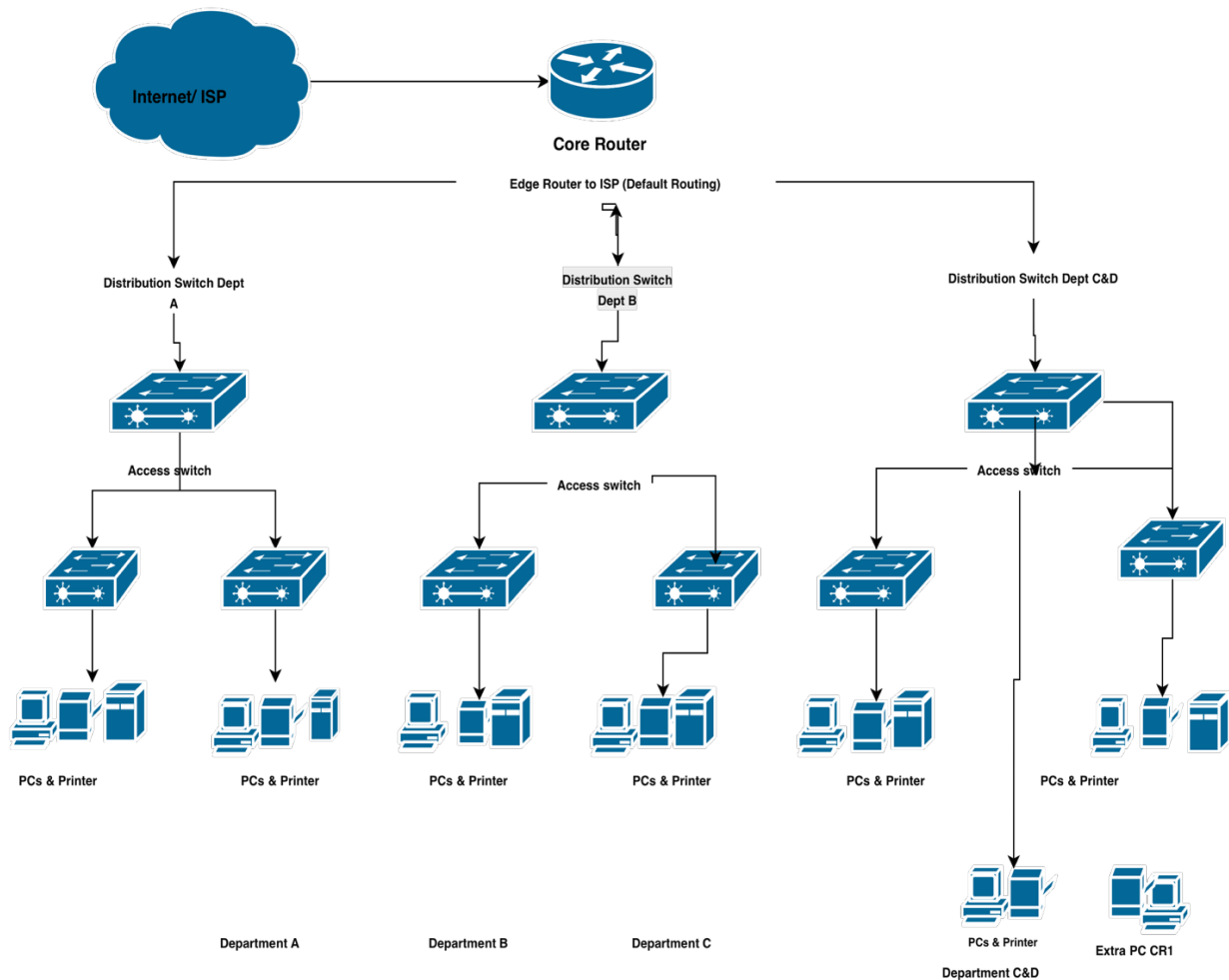
Milestone 1 Document

1. Client Requirements

The assigned client for this project is **Motswaledi Civil & Structural Engineers (Mahikeng)**, operating in the **engineering industry**. The organisation requires a reliable, scalable, and secure computer network to support its operations. The following requirements and constraints must be addressed:

- **Assigned Addressing Block: 172.30.40.0/23** Provides 512 usable IP addresses, which must be subnetted to accommodate multiple departments and future growth.
- **Connectivity & Services:** The network must enable seamless communication between departments, support essential services (such as DHCP and DNS), and ensure successful data exchange across all nodes.
- **Design Constraint:** Legacy devices without modern security features must still be integrated safely into the network, requiring careful planning of topology and security measures.
- **Assigned Networking Challenge:** Implementation of Default Routing at the edge/ISP path. The solution must configure, verify, and demonstrate correct default routing functionality.

2. Physical Topology

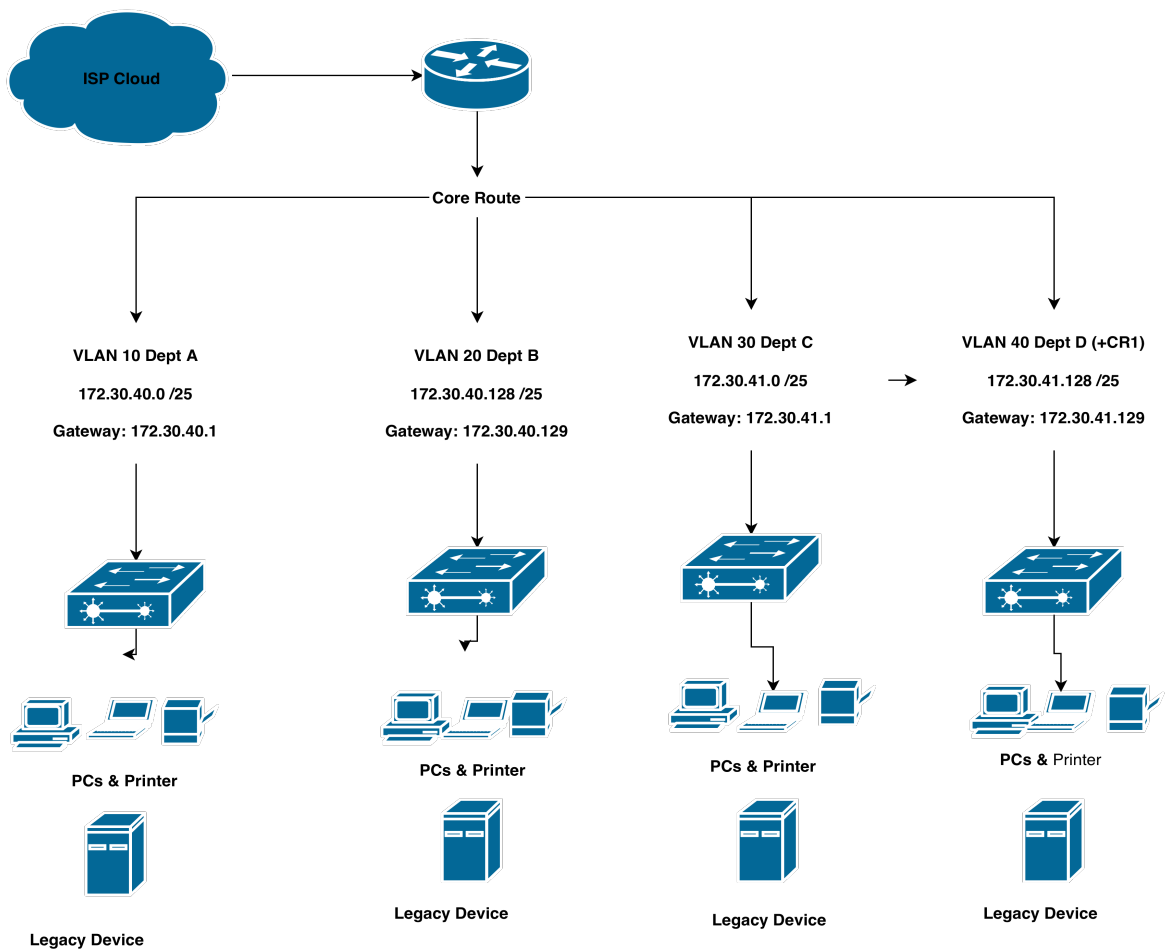


The physical topology describes the actual hardware layout and cabling of the client's network. It shows how routers, switches, and end devices are physically connected to ensure reliable communication.

- **Core Router**
 - Positioned at the edge of the network.
 - Connects directly to the ISP cloud for external communication.
 - Handles the assigned networking challenge: Default Routing.
- **Distribution Switches**
 - One per department to aggregate traffic.
 - Provide structured connectivity between the core router and departmental access switches.
 - Ensure scalability and easier troubleshooting.
- **Access Switches**
 - Connect end devices such as staff PCs, printers, and legacy systems.
 - Provide departmental connectivity at the edge of the network.
 - Allow VLAN segmentation for logical separation.
- **End Devices**
 - Staff computers, printers, and legacy devices.
 - Connected via access switches to ensure departmental communication.
 - Legacy devices are integrated safely, respecting the design constraint.

- **ISP Cloud**
 - Represents external connectivity beyond the organisation.
 - Linked to the core router to enable internet access and external communication.

3. LOGICAL DESIGN



Layer	Purpose	Example Devices
Core Layer	Manages routing and external connectivity	Core Router
Distribution Layer	Controls inter-departmental traffic	Departmental Switches
Access Layer	Connects end devices and assigns VLANs	PCs, Printers, Legacy Devices

Department	VLAN ID	Subnet	Gateway
Dept A	VLAN 10	172.30.40.0/25	172.30.40.1
Dept B	VLAN 20	172.30.40.128/25	172.30.40.129
Dept C	VLAN 30	172.30.41.0/25	172.30.41.1
Dept D (+CR1)	VLAN 40	172.30.41.128/25	172.30.41.129

4. IP Addressing Plan

The assigned addressing block for the client is **172.30.40.0/23**, which provides 512 usable IP addresses. This block has been subnetted to accommodate departmental needs, ensure scalability, and integrate the client's change request (CR1) without requiring a full redesign.

Subnet Allocation

Department	Subnet	Usable Hosts	Gateway	Broadcast
Dept A	172.30.40.0/25	126	172.30.40.1	172.30.40.127
Dept B	172.30.40.128/25	126	172.30.40.129	172.30.40.255
Dept C	172.30.41.0/25	126	172.30.41.1	172.30.41.127
Dept D (+CR1)	172.30.41.128/25	126	172.30.41.129	172.30.41.255

Addressing Strategy

- **Gateway Addresses:** The first usable IP in each subnet is reserved as the default gateway.
- **DHCP Allocation:** Remaining usable IPs are dynamically assigned to staff PCs and printers.
- **Legacy Devices:** Assigned static IPs within each subnet for stability and monitoring.
- **CR1 Expansion:** Dept D's subnet reserves 8 additional IPs for new staff, ensuring seamless integration without redesign.
- **Scalability:** Each subnet supports up to 126 hosts, leaving room for future growth.

5. Github

<https://github.com/mazibukothabanggift/CMPG-325-Individual-Semester-Project.git>

